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Subsidy policy optimization of multimodal transport on emission reduction considering carrier pricing game and shipping resilience: A case study of Shanghai port

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ABSTRACT

To save energy and reduce carbon emissions, subsidy policies have been implemented to improve the competitiveness of railway and waterway transport in the multimodal transport system of port shipping. However, there is a lack of systematic research on the assessment of subsidy distribution with the consideration of shipping resilience and emission reduction. Disordered competition hinders the development of the multimodal transport market and also deters the achievement of carbon neutrality. To address these issues, this study proposes an optimized subsidy strategy with a bi-level subsidy decision-making model considering the pricing game between various carriers. Specifically, the amount of government subsidies is substituted into the upper model, and the carrier's pricing decision is obtained by solving the Nash equilibrium of the tripartite pricing game. An activity-based approach is used to calculate the emission reduction corresponding to the subsidy amount. The variance of the transport time is used to measure the resilience of transportation. Finally, the optimal subsidy strategy is obtained through quantitative comparison. This study employs the multimodal transport of containers based on Shanghai Yangshan Port as a case study. The sensitivity analysis discloses that the time value of goods and freight market conditions should raise more attention in the process of subsidy policy formulation. Moreover, the waterway carrier should try to improve resilience and reduce the risk of delay. This study innovatively considers the pricing game between highway, railway, and waterway carriers and provides a rational basis for the formulation of subsidy policies in multimodal transport through quantitative analysis.

1. Introduction

In 2018, the transportation industry was the second largest contributor to carbon emissions, accounting for over 30% of the total emissions (Wang et al., 2020). As one of the most important carbon emission sources in the transportation field, road-based freight transport contributes to 79% of carbon emissions. Taking container transport in China as an example, the carbon emission of road transport is 5.7 times that of railway transport (Huang and Lin, 2011). Xu et al. (2023) demonstrated that during the COVID-19 pandemic, carbon emissions decrease with an increase in newly confirmed cases, but they increase with port congestion, waiting times, and container idle rates. Gu and Liu (2023) also found that the COVID-19 pandemic has worsened port congestion issues, leading to increased air pollution. Therefore, ports need to improve operational energy efficiency, alleviate congestion, and reduce waiting times. In addition to studying ship scheduling issues to reduce emissions (Chen et al., 2022a,b), the development of multimodal transportation is also a viable solution. Rail and water transportation have higher energy efficiency compared to road transport. Moreover, they also have effectively address port congestion issues caused by excessive trucking.

In addition to the significant carbon emissions, the preference for road transport increases pressures on transport infrastructure and external negative benefits, such as congestion and noise (Euchi et al., 2021). However, road transport is currently the main choice of multimodal transport, mainly because of its good timeliness, convenience,

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and quick door-to-door services. What's more, because of the high transshipment costs, the transport costs of medium and short-distance sea-rail multimodal transport are higher, and shippers always pay great attention to transport costs. Providing subsidies to help railway and waterway carriers reduce freight rates can improve their competitiveness (Santos et al., 2015).

Countries in Europe and the United States introduced policies to support the development of greener and environment-friendly multimodal transport and improve the market competitiveness of sea-rail and river-sea multimodal transport. For example, the New York New Jersey Port Authority provided a series of subsidy policies for multimodal transport, including a subsidy of $25 for each heavy container, a subsidy of $200,000 for new routes, free storage, and returning empty containers. In 2009, the Belgian government formulated the Federal Public Service Transport and Transportation Act C-2009/14189 for the development of sea-rail multimodal transport and provided subsidies to ports, transit stations, and carriers (Tang and Yang, 2016).

Meanwhile, there is a lack of quantitative assessment and evaluation of subsidy policy of multimodal transport. In general, there are two main difficulties in formulating a quantitative multimodal transport subsidy policy. The carrier expects to continue receiving subsidies and has not estimated how much carbon emissions have been reduced by multimodal transport, which is the key basis for obtaining subsidies. Besides, the multimodal transport market is highly competitive. There is much more reduction space for road transport prices (Tawfik and Limbourg, 2018) and the competitiveness of multimodal freight is sensitive to pricing strategies (Bontekoning et al., 2004). Without rational subsidies, it will be impractical to achieve a more energy-saving and environment-friendly transport mode.

To enhance the market competitiveness of port-based multimodal transport and better achieve the goal of acceptable resilience and emission reduction, this study conducts quantitative research on modeling and evaluating multimodal transport subsidy policies. This study adopts the following method to achieve the above objectives. A bi-level subsidy decision-making model is established. The upper-level model is a tripartite game model, in which the pricing decision of highway, railway, and waterway carriers is obtained through the graphical method. The lower-level model is the logit model. With the pricing decision derived from the upper-level model, the proportion of the shipper's selection of transportation mode can be assessed. With the activity-based method, the carbon emissions of the three transportation modes are estimated. Resilience is the ability of a transport system to maintain normal operations despite a certain level of disruption. Multimodal transportation systems are more complex than other transportation systems, so it is necessary to consider the ability to operate when a certain step is disturbed. The variance of the transport time is used to measure the resilience of transport. Then the optimal subsidy strategy is obtained by comparison. The activity-based approach is used to estimate the carbon emission reductions from transportation modal shifts after railway and waterway carriers have received government subsidies.

This study innovatively considers the pricing game among highway, railway, and waterway carriers and aims to maximize the carbon emission reduction per unit subsidy. The risk of delay is also taken into account when calculating the time cost of transportation. After verifying the effectiveness of the model, this study proposes a quantitative multimodal transport subsidy strategy with a case study. Shanghai Yangshan Port is primarily studied involving multimodal container transport in this study. The modeling results and proposed evaluation model can be adapted to multimodal transport in other ports and shipping routes.

This paper is organized as follows. Section 2 presents a literature review related to the multimodal transport optimization model, multimodal transport subsidies, and games in multimodal transport. Section 3 establishes a bi-level subsidy decision-making model. In Section 4, the optimal subsidy strategy is obtained by substituting data, and the results are analyzed and discussed. The final section demonstrates the conclusion of this study.

2. Literature review

As an emerging research area, multimodal transportation has gained increasing research interest over the past 20 years (Caris et al., 2008, 2013). Most scholars build route optimization models from the perspective of carriers. Compared with unimodal transport systems, multimodal transport systems which involve more factors are more complex, so one or several factors in multimodal transport systems are usually selected for the study. For example, Hao and Yue (2016) proposed an optimization model based on dynamic programming to minimize the generalized transportation cost. Guo et al. (2021a,b) employed the Sichuan-Tibet region as an example to study the risk propagation law of multimodal transportation and proposed methods to control the risk of multimodal transportation. Some scholars have also developed multi-objective optimization models considering time cost, risk, etc. Alumur et al. (2012) proposed a linear mixed integer programming model to analyze the trade-off between cost-effective routes and transportation time constraints using the example of multimodal transport in Turkey. The siting strategy of emergency relief facilities in a multimodal transportation network is studied to improve the resilience of this network (Guo et al., 2021a,b).

More recently, the effect of energy saving and emission reduction of multimodal transportation has been paid attention to by many scholars. For example, Janic (2007) established a multimodal freight network model combining internal and external costs. Yang et al. (2021) developed models to simultaneously optimize shipping network design policies and tariff policies to minimize systemic CO2 emissions. Sun and Lang (2015) used Pareto optimal to select the optimal multimodal route and transformed transportation costs, time costs, and carbon emission costs into a single objective function to solve. A genetic algorithm is designed to study a container shipping network design problem, which takes both the carbon tax and shipper's choice inertia into account (Gao et al., 2022). Ziaei and Jabbarzadeh (2021) proposed a multi-objective model based on the previous work, considering multiple uncertainties, while minimizing carbon emissions, risk, and transportation costs. A dynamic calculation method of multimodal transport line parameters based on "Witness" software simulation is proposed to reduce carbon emission, and the China-Europe railroad network from Wuhan, China to Berlin, Germany is taken as an example (Wang et al., 2020).

The government can play a positive role in promoting the development of multimodal transport, giving full play to the characteristics of railway and waterway transport in multimodal transport with low carbon emissions, thus achieving energy saving and emission reduction. Based on the multimodal freight location-allocation model and hub location theory, Santos et al. (2015) argued that subsidies can improve the competitiveness of multimodal transport, while internalizing external costs, such as carbon taxes, can harm multimodal transport. To determine the impact of freight subsidy policies and form a reasonable freight subsidy mechanism, many researchers have also studied the impact of subsidy policies on multimodal transport development. But there are few quantitative studies among them, and most of them are qualitative studies or only discuss the difference between with and without subsidies. Researchers have argued the importance of multimodal transport subsidies from different perspectives. For example, Song (2014) demonstrated that government financial subsidy support plays a crucial role in the improvement of port container throughput. Xu et al. (2021) developed a three-party evolutionary game model incorporating prospect theory to investigate the optimal strategies of upstream and downstream governments and shipping companies. The major suggestion is that under proactive and vigilant government regulations, shipping companies are more inclined to adopt clean energy sources. Based on an integer linear optimization model, Carreira et al. (2012) pointed out that when rail transportation or transshipment costs

are subsidized by the government, external costs are reduced and the competitiveness of multimodal transport is increased. By comparing the situation with and without government subsidies, Jiang et al. (2015) established a model of multimodal transport subsidy productivity and demonstrated that government subsidies play an important role in promoting the development of multimodal transport and improving the integrated transportation system. A small number of researchers have conducted quantitative studies on multimodal transport subsidy policies. Taking multimodal transportation in the Yangtze River Delta as an example, Yin et al. (2021) solved the model based on Dijkstra's algorithm and pattern search algorithm to maximize rail freight per unit posting and minimize total cost to obtain the optimal subsidy policy and found that the proportion of rail freight increased by 20% when the proportion of subsidy increased to 30%.

The main reason why subsidies are difficult to analyze quantitatively is that multimodal transport, as a complex system, involves multiple parties in the game. For government subsidies to carriers, the amount of the subsidy is not equal to the amount of the carrier's price reduction. Some scholars have already tried to apply game theory to the study of multimodal transportation. Lu (2017) constructed and solved the game model of freight mode selection between shippers and carriers, and proposed suggestions to promote the development of multimodal transport based on the dynamic game process, proving that game theory is applicable to the analysis and study of freight mode selection. Tsao and Linh (2018) used game theory to help ports, waterless ports, and goods owners optimize their interests and guide them in determining their decision variables, demonstrating that the development of the waterless port and multimodal transport can reduce the carbon cost of road transport. Qu et al. (2020) developed a pricing model based on game theory to evaluate the impact of different incentive strategies on multimodal transport systems. A differential game model is developed to demonstrate that government incentives for ports and shipping companies can motivate both parties to provide low-carbon services (Meng et al., 2022). To optimize the shipping network design and carbon tax collection scheme, Liu et al. (2021) established an optimization model aiming at both maximizing the total profit of the alliance and improving the fairness of profit distribution. Jiang et al. (2020) developed a nested logit model to describe shippers' choice of ports, built a Stackelberg model for two ports and solved it to obtain the optimal location of ports, and pointed out that for shippers, inland transportation cost plays a crucial role in choosing which port to visit.

The current research on multimodal transport subsidies and games has achieved preliminary results, but there are still several gaps, which are mainly reflected in the following three aspects.

(1) Due to the complex influencing factors of multimodal transport and the multiparty game involved, the current research on subsidy policy is mainly limited to qualitative analysis. The design of optimization models and algorithms needs to be enriched and improved. Although researchers pointed out that internalization of external costs helped the development of green multimodal transport through the analysis of the broad cost of multimodal transport, few researchers have established a subsidy decision model with the objective of emission reduction effect.
(2) In the limited articles that analyze subsidies quantitatively, researchers have considered the game between the government, carriers, and shippers, but have not considered the game between carriers under different subsidy situations.
(3) Numerous scholars have considered the time value of goods when calculating the broad cost of multimodal transport, but have not paid attention to its resilience. Due to the complexity of multimodal systems, shipments are often delayed for various reasons, resulting in missing shipping schedules. The risk of delay in river-sea multimodal transport, therefore, needs to be calculated, but few scholars have taken this into account.

Based on these knowledge gaps, the innovation points of this study are as follows.

(1) A bi-level subsidy decision-making model is established with the objective of maximum emission reduction per unit subsidy. Through quantitative analysis, it aims to provide a comprehensive and reasonable basis for the formulation of subsidy policy.
(2) In this model, the upper model considers the pricing game among highway, railway, and waterway carriers, which is in line with the reality and the market operation law and has application value.
(3) In the lower logit model, demurrage charges resulting from the risk of delay are factored into the calculation of time costs. Furthermore, the impact of the resilience of shipping on shippers' choice preferences and carriers' operating profit is analyzed.

3. Problem definition and methodology

In general, inland goods is transported to the port through one of the following three modes generally (Huang, 2016). The first mode of transportation is to directly transport the container to the port by truck, providing door-to-door service without midway transshipment. The second mode of transportation is sea-rail multimodal transportation. The containers are transported by truck to the local railway station, and then transported by rail to the port city. If the railway is directly connected to the port, there will be no distance between the railway station and the port in the same city. Otherwise, after arriving in the port city, the container will be transported to the port by shuttle car. In the third mode of transportation, containers need to be transported by road and then by water to the port city. Similar to the second mode of transportation, the necessity of short-distance transportation depends on whether river-sea multimodal transport is realized.

This study doesn't consider the possibility of switching transportation modes during the commodity transportation based on practical situations. Specifically, China has a highly developed rail network. Carriers in cities located along rivers typically don't find it cost-effective or time-efficient to change transportation modes midway due to the additional loading and unloading procedures involved. However, for cities that are not located along rivers, it is important to consider a combination of rail and water transportation as a viable option. Fig. 1 shows the transport process of containers exported through multimodal transport.

Due to the requirement of carbon neutrality, the government aims to reduce carbon emissions and promote the development of multimodal transport by providing subsidies to railway and waterway carriers in multimodal transport. From the perspective of government, there is a desire to minimize carbon emissions with fewer subsidies. Then the objective function of this study is as follows:

$$objective\ function = max\left(\frac{E - E_0}{TotalS}\right) \tag{1}$$

Where E refers to the total carbon emissions with subsidies; E_0 refers to the total carbon emissions without subsidies; and $TotalS$ refers to the total number of government subsidies for railways and waterways. To obtain the optimal subsidy strategy, a bi-level subsidy decision-making model is established.

3.1. The upper model——tripartite game model

Multimodal transport business is mainly participated by highway, railway, and waterway carriers, which provide similar but not completely homogeneous services. As a result, the market competition is fierce and carriers often conduct pricing games to attract customers. In order to reduce carbon emissions and improve the competitiveness of railway and waterway transport, the government grants multimodal

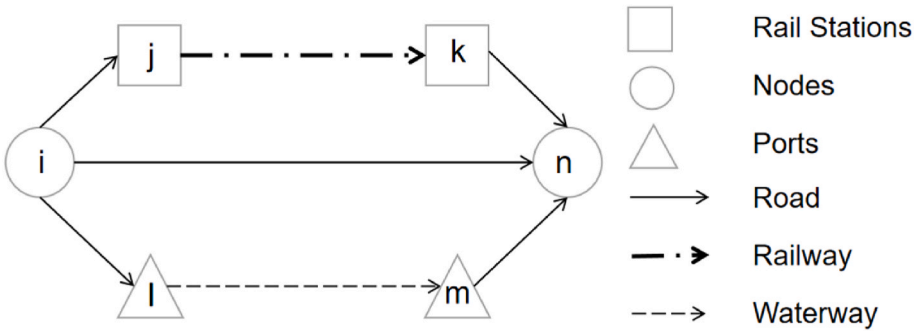


Fig. 1. The transport process of containers exported through multimodal transport.

transport subsidies to railway and waterway carriers. However, it should be noted that the actual price reduction amount in railway and waterway transportation is not necessarily equal to the subsidy amount, and the pricing of highway transportation will change accordingly, too. Therefore, a tripartite game model is established to assess the price reduction range of highway, railway, and waterway carriers when the subsidy amount is determined.

Feng and Li (2017) analyzed the relationship between freight transport price and service quality, and points out that under the condition of strong product substitutability, Nash equilibrium can be reached when both parties do not improve service quality and only conduct price competition. Therefore, in this case, the carrier pricing strategy can be obtained by solving the Nash equilibrium of the tripartite game model.

In this study, the graphical method is used to solve Nash equilibrium. In order to convert a continuous problem into a discrete problem, it is stipulated that carriers can only reduce prices in the form of discounts. The principle of the graphical method is: after all participants excluding the last one have chosen a strategy, the last participant chooses the strategy with the greatest benefit to him, and the corresponding maximum value is underlined. After listing all possible situations, the strategy with three lines is the Nash equilibrium solution. In the decision-making process, the complete information conditions required by the graphical method are met, that is, each carrier knows the price, profit, subsidy amount and game strategy set of other transportation modes, because all these information can be obtained from the company's financial reports, government bulletins and other channels.

On the premise that the subsidy is determined, the game strategy set of highway, railway, and waterway carriers can be obtained first, and a three-dimensional matrix can be listed. Each matrix unit is the profit of the three parties when the pricing decision of the three parties is known. When the price reduction strategies of the two carriers are decided, the last carrier can find the price reduction strategy that maximizes its operating profit and complete the underline. The advantage of this discrete solution method is that it is highly robust and practical with the employment of all Nash equilibrium solutions under different parameters. Besides, the difference between the solution and that obtained by the continuous optimization algorithm is no more than 1%. In other word, the algorithm achieves a high degree of accuracy. Although the method is slower when the pricing strategy is more discrete (e.g., taking a minimum price reduction of 1%) than the continuous optimization algorithm, this minor difference does not require a real-time response in practice and does not affect practicality. Based on performances in accuracy, computational cost, robustness, and practicality, the proposed method was therefore employed for this study.

To solve Nash equilibrium, this study needs to obtain the proportion of the shipper choosing three transportation modes on the premise of the carrier's pricing decision, so as to calculate the profit margin of highway, railway, and waterway carriers. To achieve this goal, it is necessary to combine the upper tripartite game model with the lower logit model.

3.2. The lower model——logit model

In the selection of transportation mode, the logit model was widely used by shippers to decide which scheme to choose according to the utility value of the transportation mode (Lee et al., 2016; Tavasszy et al., 2011). In addition to transportation price and time value, utility value also includes the reliability and convenience of different transportation modes, thus logit model has wide applicability and strong reliability. For transportation mode i, the utility value is calculated as follows:

$$U_i = V_i + \in_i \tag{2}$$

$$V_i = \theta(P_i + T_i + A_i) \tag{3}$$

Here V_i refers to the deterministic utility. $\in_i$ refers to the stochastic utility. P_i is the total transportation cost, T_i is the time cost, and A_i is the compensation cost for the convenience and reliability of the transportation scheme. The lower the compensation cost, the worse the safety and convenience. $i = 1$, 2, and 3 represent highway, railway, and waterway transport, respectively. P, T, and A form the broad transportation cost together. θ refers to the total cost factor, which reflects the relationship between broad transportation costs and utility values. Moreover, McFadden (1973) proved that when $\in_i$ conforms to the double exponential distribution, the selection ratio of different transportation modes can be expressed as:

$$probability_i = \frac{e^{V_i}}{\sum e^{V_i}} \tag{4}$$

This is the calculation formula of the logit model. To calculate the selection proportion of different transportation modes, P_i and T_i are calculated respectively and substituted into the logit model.

3.2.1. Price analysis of transportation

The prices of highway transport mainly include basic freight, highway toll fee, loading and unloading fee (*luf*), and container fee paid each time a container is used (*cf*). The transport price of a single container is as follows.

$$P_{road} = (c_{road} + \gamma_1) * d_{road} + luf_{road} + cf \tag{5}$$

Where P_{road} refers to the total price of container road transport, c_{road} refers to the coefficient of basic freight, γ_1 refers to the coefficient of highway toll fee, and d_{road} refers to the distance of road transportation. Since road transportation is door-to-door, there is no transshipment cost.

The prices of railway transport consist of fixed costs and variable costs. The transport price of a single container is as follows.

$$P_{rail} = (c_{rail} + \gamma_2) * d_{rail} + p + luf_{rail} + q_1 + q_2 \tag{6}$$

Where P_{rail} is the total price of container railway transport. Variable costs mainly include coefficient of basic fright (c_{rail}) and railway construction fund (γ_2), while fixed costs mainly include base

price (p) and loading and unloading fee (luf_{rail}). Other costs such as electrified railway surcharge can be ignored, since they account for a small proportion of the cost. d_{rail} refers to the distance of rail transportation. q_1, q_2 refers to the transshipment costs at both ends. If the railway is directly connected to the port, $q_2 = 0$.

The prices of waterway transport mainly include inland waterway transport costs and loading and unloading fee (luf_{ship}). The transport price of a single container is as follows.

$$P_{ship} = c_{ship} * d_{ship} + luf_{ship} + q_3 + q_4 \tag{7}$$

Where P_{ship} refers to the total price of container waterway shipping transport, c_{ship} refers to the coefficient of waterway freight, d_{ship} refers to the distance of waterway shipping transportation, and q_3, q_4 refers to the transshipment costs at both ends. If the container can be directly transported to the port by water from the ports in inland cities, $q_4 = 0$.

3.2.2. Time cost

Different transportation modes have different transportation duration and corresponding time cost, which will affect the shipper's choice of transportation mode. Therefore, the time cost is an important part of the logit model, which mainly includes the time value of goods and the cost of renting containers. The time cost of a single container is as follows.

$$T = tv * t + dc \tag{8}$$

$$t = t_1 + t_2 + t_3 = \sum \frac{d}{v} + t_2 + n * t_0 \tag{9}$$

$$cd = \frac{1}{ap}\int_{x=0}^{ap} p_r * \left\{ max[(ap - x - fd), 0] * \Phi\left(\frac{x - (ap - \bar{t})}{\sigma}\right) + max[(2 * ap - x - fd), 0] * \left[\Phi\left(\frac{2 * ap - x - \bar{t}}{\sigma}\right) - \Phi\left(\frac{ap - x - \bar{t}}{\sigma}\right)\right] + max[(3 * ap - x - fd), 0] * \left[\Phi\left(\frac{3 * ap - x - \bar{t}}{\sigma}\right) - \Phi\left(\frac{2 * ap - x - \bar{t}}{\sigma}\right)\right] + \ldots \right\} \mathrm{dx} \tag{10}$$

Where T is the time cost of container transportation, tv is the time value of goods, t is the total time, which is composed of transport time (t_1), waiting time (t_2), and transit time (t_3), and dc is the demurrage charge, which is related to resilience. t_1 is calculated from the transportation distance and transportation speed. t_2 is the waiting time before the container is transported and before it is transferred. As trains and ships can only leave after all containers are loaded and unloaded, t_3 is the loading and unloading time t_0 of a single container multiplied by the traffic volume n of the vehicle. In reality, the shippers of goods with high unit time value are more inclined to choose highway and railway transportation with shorter transportation time, while the shippers of goods with low unit time value prefer waterway transportation. Since this study considers multimodal transport of all containers from a specific city to a specific port, tv is the average unit time value of the goods at the port.

When exporting, if the container cannot be loaded onto the vessel within free time (ft), the consignee will have to pay dc. Since ships have a relatively fixed arrival period (ap), even if the delay is not long, dc may be high. In this study, it is assumed that the number of days between the date of container shipment and the arrival date of the next ship follows a mean distribution ($x \sim U(0, ap)$). Although weekend and holiday shipments differ from weekdays, the probability of each day in a ship period being a weekend or holiday becomes equal when the sample size is large enough. Therefore, for any given container, the probability of starting a shipment on each day of the shipping period is nearly equal. Based on the data statistics of Chen et al. (2018), this study assumes that the total transit time approximately follows a normal distribution ($t \sim N(\bar{t}, \sigma)$). In cases where the exact probability density or distribution of sailing time is known, the method proposed in this study can still be adapted to estimate the delay cost in the future. p_r is the daily fee to be paid after the overdue period, which is basically the same for different transportation modes. dc is calculated by multiplying the probability density of x, p_r, the number of days in which dc is paid, and the probability of that occurrence.

3.3. Combination of upper model and lower model

In the tripartite game model, the carrier needs to choose the pricing decision that maximizes its profit when other carriers' pricing decisions are known. This goal can be achieved by bringing the price reduction ratio (D) of the three carriers into the logit model to obtain the profit (R) corresponding to the pricing decision. According to the principle of the graphical method, the Nash equilibrium of the tripartite game, which determines the pricing decisions of highway, railway, and waterway carriers, can be found by using the for loop.

The determined D *is* substituted into the logit model to obtain the deterministic utility (V) and the selected proportion(pro) of different transportation modes.

$$V_i^{'} = \theta\left[(1 - Di) * P_i + T_i + A_i\right] \tag{11}$$

$$pro_i^{'} = \frac{e^{V_i^{'}}}{\sum e^{V_i^{'}}} \tag{12}$$

Given the cost of container transportation (C) and the amount of government subsidy (S), the profit rate (r) and profit (R) of highway, railway, and waterway carriers after price reduction can be calculated. i = 1, 2, and 3 represent highway, railway, and waterway transport, respectively.

$$r_i = \frac{(1 - Di) * P_i + S_i - C_i}{C_i} \tag{13}$$

$$R_i = r_i * pro_i^{'} \tag{14}$$

Then, for each carrier, it is only necessary to find the best price reduction ratio that maximizes its operating profit. However, for a tripartite game model with a large number of decisions and three game players, it is not necessarily possible to find the only Nash equilibrium solution. There may be no Nash equilibrium solution or multiple Nash equilibrium solutions. How to solve this problem will be mentioned in the case analysis.

This study establishes the bi-level model with two motivations. Firstly, this model adheres to the definition of a bi-level model. The decision-makers in the tripartite game model are evidently the carriers. The decision-makers in the Logit model are the shippers who choose the mode of transportation based on the upper-layer decision (e.g., pricing). Each shipper can make decisions within a certain range and the Logit model estimates the proportion of choices made by all shippers when the upper-layer decision is determined. Secondly, the Logit model is a commonly used discrete choice model. For example, in transportation mode choice studies (Lee et al., 2016; Tavasszy et al., 2011), the transportation prices and time costs of each mode need to be calculated separately.

3.4. Calculation of carbon emission

In order to obtain the optimal subsidy strategy, it is also necessary to calculate the carbon emissions before and after the subsidy. McKinnon and Piecyk (2010) pointed out that two methods to estimate carbon dioxide emissions in freight operations were used frequently. The first is to record energy use and convert it into carbon dioxide emissions. The second method is the activity-based approach, but it is difficult to determine the carbon emission factors. Qu et al. (2016) established the first multimodal transport model considering transport costs, transit

costs and external costs at the same time, and used the activity-based approach to calculate carbon emissions, pointing out that this calculation method avoids elements that are difficult to measure or calculate, and is applicable to different transport modes. Jiang et al. (2012) took Shanghai Port as an example, estimated and compared the carbon emissions of roads, railways and waterways in multimodal transport respectively using the activity-based approach. Therefore, this study uses the activity-based approach to estimate carbon emissions, and the calculation formula is as follows. i, j = 1, 2, and 3 represent highway, railway, and waterway transport, respectively.

$$o_i = w * \left(d_i * f_i + \sum_j tr_{ij} * f_j + \sum ce \right) \tag{15}$$

$$e = pro * o^{'} \tag{16}$$

Where o refers to the total carbon emission of transporting a container, w refers to the weight of a container, d refers to the transportation distance, ce is the carbon emission from the loading and unloading process in kg/t, and f refers to the carbon emission factor in kg/(t*km). This study also considers the carbon emissions in the transport process. tr_{ij} is the total transit distance for the jth transport mode in the ith transport scheme. The transit distance for road transportation is 0. After the selection proportion of three transportation modes is obtained, the average carbon emission of a single container (e) can be calculated through matrix multiplication. For the model proposed in this study, the total container traffic volume (N) of a city to a specific port is not necessary, because it will be omitted in the calculation without affecting the calculation results. The calculation process of the value of the objective function (F) is as follows.

$$\begin{aligned} F &= max\left(\frac{E - E_0}{TotalS}\right) \\ &= max\left[\frac{(e - e_0) * N}{pro * S^{'} * N}\right] \\ &= max\left(\frac{e - e_0}{pro * S^{'}}\right) \end{aligned} \tag{17}$$

Where E_0 and E are the total carbon emissions before and after subsidy, and e_0 and e are the average carbon emissions of each container before and after being subsidized.

3.5. Algorithm

The optimal subsidy strategy is found by combining the upper logit model with the lower tripartite game model. In the calculation process, a double cycle is designed. The upper cycle is the government subsidy for railway carriers, and the lower cycle is the government subsidy for waterway carriers. Then all subsidy strategies for carriers can be exhausted, and the government subsidy policy that maximizes the value of the objective function (F) can be found.

In this study, it is stipulated that the subsidies will be increased by 10 yuan each time, and the maximum amount will not exceed 50% of the original price, which conforms to reality. The algorithm is implemented by using Matlab programming, and the operation steps are as follows.

Step 1: Use the formula in Section 3.2.1 and Section 3.2.2 to calculate the transportation cost and time cost of the three modes.
Step 2: Through the tripartite game model, calculate the price reduction ratio of highway, railway, and waterway carriers without government subsidies (D_0)
Step 3: Substitute D_0 into the logit model to obtain the selection proportion of different transportation modes (pro_0)
Step 4: Calculate the average carbon emissions of a single container without subsidies (e_0) using the activity-based method
Step 5: Substitute the subsidies (S) into the tripartite game model to obtain the price reduction ratio (D) of highway, railway, and waterway carriers
Step 6: Substitute D into the logit model to obtain the selection proportion of different transportation modes
Step 7: Calculate the average carbon emissions of a single container with subsidies (e) using the activity-based method
Step 8: Calculate F through equation 17
Step 9: Repeat Step 5 to Step 8 to try all subsidy strategies, and obtain the maximum of F and the corresponding optimal subsidy strategy through comparison

4. Case study

4.1. Case background

In China, in terms of geographic location and economic development role, Shanghai is a bridgehead to promote the implementation of national strategies. With the rapid development of ports worldwide, the ecological and environmental pollution in port cities, such as Shanghai, has become increasingly severe. Finding ways to balance the port industry with the urban environment is an urgent issue that needs to be addressed. Chen et al. (2022a,b) discovered that although the coupling and coordination between the economic development of Shanghai Port and the urban ecological environment have shown an overall upward trend, there is still significant room for improvement, and it is necessary for Shanghai Port to enhance container transportation, promote energy conservation and emissions reduction, and vigorously promote the development of green ports. As the world's largest container throughput port and the world's largest fully automated port, Shanghai Yangshan Port has a strong potential for the development of multimodal transport. But at present, due to preliminary planning problems, the train cannot go straight to the port. Because of the high cost of transshipment, the costs of sea-rail multimodal transport are considerable and the proportion of sea-rail multimodal transport is less than one percent. To promote the development of sea-rail multimodal transport, the Shanghai Government provides a subsidy of 600 yuan per container for railway carriers recently. However, the gradual withdrawal of the allocation is considered. Carriers expect to continue receiving subsidies without estimating how much carbon emissions are reduced by multimodal operations, which is a key basis for receiving subsidies. Therefore, the choice of Shanghai Yangshan Port as the case study to propose a scientific solution for government subsidy policy and calculate the corresponding carbon emission reductions is of practical significance to promote the development of multimodal transport.

Shanghai is located in the Yangtze River Delta with convenient transportation. Containers from all over China are mainly transported to Yangshan Port for export by road, rail, and waterway. As the capital of Jiangsu Province, an important industrial city and economic center in the lower reaches of the Yangtze River, Nanjing is one of the regional center cities in the economically developed area of the eastern coast of China and also services as the hub city in the core economic area of Yangtze River Delta City Cluster. Therefore, it is representative to study the multimodal transportation of containers from Nanjing to Shanghai Yangshan Port for export.

There are three main routes for transporting containers from Nanjing to Shanghai Yangshan Port. The first option is road transport, where containers are transported to Yangshan Port via the Shanghai-Nanjing Expressway. Approximately 58% of shippers choose road transportation. The second option is river-sea multimodal transport, where containers are transported from the factory to Longtan Port by truck and then directly to Yangshan Port via the Yangtze River, China's golden waterway. The Nanjing government has long attached importance to the development of water transport, providing subsidies for all aspects of shipping, resulting in 39% of carriers choosing river-sea multimodal transport. The third option is sea-rail multimodal transport. After the

containers are shortly barged from the factory to Yaohuamen Station by truck, they are transported to Luchao Port Container Center Station by Shanghai-Nanjing Railway and then transported to Yangshan Port by truck, for the train cannot reach Yangshan Port directly. The detailed route is shown on the map below.

4.2. Fundamental assumptions

Before using the bi-level subsidy decision-making model, several basic assumptions are proposed.

4.2.1. Fundamental assumptions of tripartite game model

(1) There is only one decision-making subject for each transportation mode.

The basic condition for the application of the tripartite game model is that each game player only has one decision-making subject. Railway transport and waterway transport services are generally provided by a specific company, such as Nanjing Shinkansen Logistics Co., Ltd., which provides railway container transport services, and Jiangsu Port Group Container Company, which provides waterway container transport services. Short distance highway transportation at both ends can be regarded as door-to-door service provided by the carrier. For example, transshipment at both ends of the railway is mainly coordinated by Shanghai Port Sea-rail multimodal Transport Co., Ltd. The actual pricing decision is made by Nanjing Shinkansen Logistics Co., Ltd., so the impact of the price change of short distance transportation on the pricing decision can be ignored. For road transport, although many small logistics companies provide container transport services, only one company will participate in each transport. Therefore, only one company providing the lowest price and the best service can participate in the game process with railway and waterway carriers.

(2) The carriers' goal is to maximize profits.

As a transportation company, the carrier provides convenient transportation services for the shipper in order to stabilize and increase customer groups, thereby increasing income. At the same time, the carrier needs to face fierce market competition, and sometimes it needs to attract customers by reducing prices. Its purpose is to improve profit margins by making small profits but quick turnover. Meanwhile, this assumption also stipulates that carriers cannot operate at a loss, which limits the number of decisions made by all parties in the game. It should be noticed that neither the carrier nor the shipper will take external costs, such as carbon emissions, as a mark that affects their own decisions. The government can link the external cost with the carrier's operating profit by means of subsidies or taxes to affect its pricing decisions. Suppose the subsidies per container are $S = [S_1, S_2, S_3]$. Since there is no subsidy for the highway carrier, S_1 keeps zero. The following constraints must be satisfied.

$$S + (1 - D) * P > C \tag{18}$$

4.2.2. Fundamental assumptions of logit model

(1) During transportation, the containers are treated as a whole and cannot be separated.
(2) All goods are transported in forty-foot long containers.

Under normal circumstances, after being packed in the factory, the containers will be transported to the destination as they are and will not be repackaged midway, which also conforms to the definition that multimodal transport can reach the destination only through "one declaration, one inspection and one release".

At present, twenty-foot long containers and forty-foot long containers are widely used, but forty-foot long containers are mainly used for sea transportation. Therefore, in this study, only forty-foot long containers are considered.

4.3. Data input

The usage of each node in Fig. 1 in this case is listed in Table 1. Since the river-sea multimodal transport can be seamlessly connected, both m and n represent Shanghai Yangshan Port, and $d_{mn} = 0$.

Additional statistics were obtained through field research and review of existing literature, as shown in Table 2.

(1) The data required to calculate the transport prices of the three modes (P) is provided by Nanjing Shinkansen Logistics Co., Ltd.
(2) The average speed of road, railroad and waterway transportation (v) is 50, 40 and 20 (km/h), respectively. A train generally carries 50 containers, and the capacity of Yangtze River freighter is generally 125 FEU. The timeliness requirements of ocean-going export goods to Shanghai Yangshan Port are relatively low, and the time value of goods (tv) is 15 (yuan/hour) (Huang, 2016). According to survey, both free time (ft) and arrival period of ships from Shanghai Yangshan Port to specific cities (ap) is about 7 days. Due to the operation of the five-scheduled train, there is less risk of delay in transporting containers by rail. Since there is no need for transshipment, there is almost no risk of delay in road transportation, so this study only considers the delay risk of waterway transportation. The standard deviation of single voyage time of the Yangtze River container ship is 1.45 (Chen et al., 2018). Although the sailing starting point is different, the standard deviation of sailing time is not a major factor in the risk of delay, so the statistic can be used.
(3) The other parameters needed to solve the logit model. The value of the total cost factor θ depends on the freight distance, and the distance from Nanjing to Shanghai Yangshan Port is relatively close with $\theta = -0.0045$ (Tavasszy et al., 2011), while the value of the compensation cost A depends on the convenience and reliability of the transportation scheme. According to the market research, the convenience and reliability of road transportation is stronger with $A = 0$, while the convenience and reliability of sea-rail and river-sea multimodal transportation are slightly weaker with $A = 100$ (Huang, 2016).
(4) By reviewing information and conducting market research, the profit rate (r) and the cost of container transportation (C) are obtained. The railway and waterway systems are so huge that the profits are difficult to measure, so the annual reports of the carrier companies are referred to. Daqin Railway Co. has annual revenue of 75.76 billion and net profit of 11.2 billion, and the profit rate of railway transportation (r_2) is 14.8%. Jiangsu Port Group Co. has annual revenue of 21.98 billion and cost of 21.05 billion, and the profit rate of waterway transportation (r_3) is 4.2%. The profit rate of highway transportation (r_1) cannot be used the profit rate of a freight company, because according to the market research, the pricing freedom of road transportation is very high. As long as a single transport is profitable, shippers will reduce prices to attract shippers, so the cost of road transport needs to be estimated to calculate r_1. Road transport costs are mainly composed of container handling charges (75 yuan), labor costs, the bridge

Table 1
Meaning of nodes.

i	Factory in Nanjing
j	Yaohuamen railway station
k	Luchao port container terminal
l	Longtan port
m	Shanghai Yangshan Port
n	Shanghai Yangshan Port

Table 2
Related parameters.

Transport distance (Unit: km)						
d_{ij}	d_{jk}	d_{kn}	d_{in}	d_{il}	d_{lm}	d_{mn}
20	402	42	430	30	423	0

Calculation of time cost						
	t1 (h)	*t2 (h)*	*t3 (h)*	*t (h)*	σ	*tv (yuan/h)*
road	8.6	1	0	9.6	0	15
rail	11.29	12	0.04*50	27.3	0	15
river	21.55	48	0.03*125	74.3	1.45	15

Price of road transportation			
C_{road} *(yuan/km)*	γ_1 *(yuan/km)*	luf_{road} *(yuan)*	*cf (yuan)*
5.2	1.94	140	25

Price of railway transportation					
C_{rail} *(yuan/km)*	γ_2 *(yuan/km)*	*p (yuan)*	luf_{rail} *(yuan)*	q_1 *(yuan)*	q_2 *(yuan)*
3.357	1.122	532	292.5	600	600

Price of waterway transportation			
C_{ship} *(yuan/km)*	luf_{ship} *(yuan)*	q_3 *(yuan)*	q_4 *(yuan)*
0.9	400	300	0

Carbon emission factors (Unit of f: kg/(t*km), unit of ce:(kg/t))				
f_1	f_2	f_3	*ce(road-rail)*	*ce(road-river)*
0.04975	0.00841	0.01733	0.27	0.31

Transit distance (Unit: km)					
tr_{21}	tr_{22}	tr_{23}	tr_{31}	tr_{32}	tr_{33}
62	0	0	20	0	0

crossing fee (150 yuan), fuel costs, highway fees. Truck driver's salary is 6000 yuan per month. Assuming one shipment per working day, the labor cost is 300 yuan. It is calculated that the fuel cost is 2.2 yuan per km. For the road distance from Nanjing to Shanghai is 430 km, the fuel cost is 946 yuan. Highway rate is 1.94 yuan per km. Taking the Beijing-Shanghai Expressway, the highway mileage is about 275 km long, then the highway fee is 533.5 yuan. Adding these costs together yields a road transportation cost of approximately 2229.5 yuan. It is known that the price of road transportation from Nanjing to Shanghai Yangshan Port is 3000 yuan, so r_1 is calculated to be 34.6%. The double-layer subsidy decision model can be solved by substituting the above data.

(5) To calculate the value of the objective function (F) to determine the optimal subsidy strategy, the carbon emission factors for the three modes of transportation (f) and the carbon emission from the loading and unloading process (ce) are also needed, as shown in Table 2 (Mao, 2013; Yu and Liu, 2018). As both road transport and transshipment processes use diesel vehicles of 20 tons or more to carry forty-foot containers, f can take the same value according to the "Bulletin of the Main Data of the National Special Survey on Road and Waterway Traffic". The unit carbon emission of inland water transport is only 1/3 of the energy consumption of road freight, and the unit carbon emission of rail transport is only 1/6 of the energy consumption of road freight, which proves that the emission reduction effect of road to rail and road to water is significant. In this case, there is no transshipment process for road transport, while sea-rail and river-river intermodal transport have two separate transshipment and loading processes with road. The carbon emissions corresponding to the subsidy strategy are calculated through equation (15), and then F is calculated through equation (17). Finally, the optimal subsidy strategy is obtained by applying the algorithm in Section 3.5.

4.4. Results

Quantitative results were obtained via substituting the above data of case study into the bi-level subsidy decision-making model. According to survey, the current rate of fare reduction will not exceed 50%, and the maximum rate of fare reduction for the three modes of transportation is set as $D_{\max} = 0.5$. First, the robustness of the model is ensured, and whether there are two or more Nash equilibrium solutions or no Nash equilibrium solutions in the lower tripartite game model is checked. After the operation, it is shown that there are only two Nash equilibrium solutions in a small number of subsidized cases, and most of the remaining cases have a unique Nash equilibrium solution. Taking $S = [0, 0, 380]$ *(yuan/FEU)* as an example, the output of the price reduction ratio (D), the profit (R), and the average carbon emissions of each container (e) for the three transportation modes is shown in Table 3.

It is found that all cases where there are two Nash equilibrium solutions have the same law. Further price reductions of the highway carrier in the second case lead to an increase in the proportion choosing road transport and an increase in carbon emissions, while at this time the profits of all carriers decrease due to price competition. This is neither in the interest of the government nor in the interest of carriers, so the second Nash equilibrium will not be reached when all parties are able to negotiate. As a result, only the first Nash equilibrium needs to be considered.

After that, the real subsidy situation is substituted to verify the validity of the model. The inability of trains to go straight to Shanghai Yangshan Port is an important factor restricting the development of sea-rail multimodal transport in Shanghai. In order to promote the development of sea-rail multimodal transport, the Shanghai government has introduced a subsidy policy to reimburse 600 yuan for the transshipment costs of containers shipped to Shanghai Yangshan port via sea-rail multimodal transport. Substituting $S = [0, 600, 0]$ *(yuan/FEU)* into the model, $D = [0.19, 0.24, 0]$, and $pro = [0.561, 0.029, 0.410]$ is obtained. And according to the prediction, the transportation mode chosen for containers transported from Nanjing to Shanghai Yangshan Port in 2022 is about 58% by road, 4% by rail and 38% by waterway, which is close to the model results. And without government subsidies, which means $S = [0, 0, 0]$ *(yuan/FEU)*, the proportion of shippers choosing sea-rail multimodal transport is only 0.23%, which is consistent with the current situation of Waigaoqiao port, another port located on the southern shore of the Yangtze River Estuary South Channel in Shanghai, which has no government subsidy and almost no sea-rail multimodal transport business.

The results in Fig. 2 indicate that the best subsidy policy is to subsidize 30 yuan per container for the railway carrier. In this case, the price of highway and railway transport is reduced by 19% and 10% respectively, the proportion of road, rail, and water transport being chosen respectively is 57.64%, 0.27%, and 42.09%, and the subsidies needed to reduce one ton of carbon emissions are 800.7 yuan. Fig. 2 also verifies the finding that the marginal utility of the government's sustainable development policies gradually decreases (Yang et al., 2021). The relationship between the number of government subsidies and the choice ratio of rail transportation and the total amount of emission reduction can also be obtained.

As shown in Fig. 3, the proportion of railway selected rises gradually as government subsidies to the railway carrier increase, but it decreases continuously as government subsidies to the waterway carrier increase when subsidies to the railway carrier remain unchanged. The reason for the stepped pattern is that carriers' decisions to reduce their prices are

Table 3
Example of two Nash equilibrium solutions.

	D	e	R_1	R_2	R_3
Case 1	[0.19 0.09 0.02]	407.9336	0.1293	0.0001	0.1261
Case 2	[0.20 0.09 0.05]	413.9922	0.1252	0.0001	0.1164

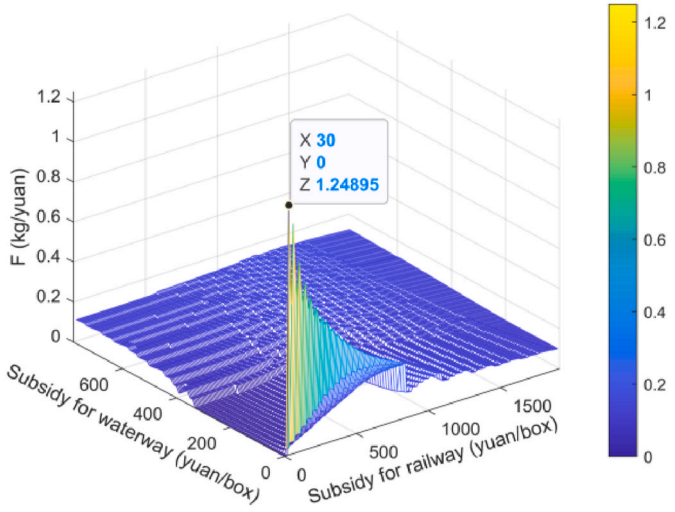


Fig. 2. The relationship between F and subsidies.

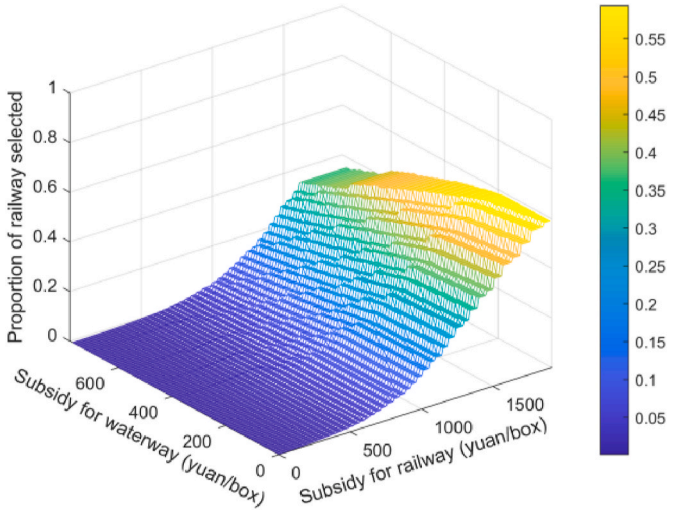


Fig. 3. The relationship between proportion of railway selected and subsidies.

not continuous and only adjust when subsidies increase to a certain level or when other carriers receiving subsidies change their pricing decisions.

Fig. 4 shows the relationship between government subsidy strategies and total emission reductions. Subsidies for both sea-rail and river-sea multimodal transport have an emission reduction effect, so the government can use this model to determine the subsidy policy according to the development objectives of multimodal transport. The model can include restrictions on the minimum selection ratio, minimum total emission reduction, and maximum subsidies to develop a more targeted subsidy strategy based on the actual needs.

Since the proportion of sea-rail multimodal transport being selected without subsidies is only 0.23%, the value of the objective function will be extremely large as long as a little subsidy is provided to the railway carrier, and the actual total emission reduction corresponding to the best subsidy scheme shown in Fig. 2 is very small. In this case, the development target of Shanghai port that the ratio of river-sea multimodal transport is not less than 52% and the average annual growth rate of container sea-rail multimodal transport volume is more than 15% will not be reached (Shanghai Municipal People's Government, 2022). To address this issue, two corresponding constraints are added to meet the multimodal transport development target proposed by the Shanghai Municipal People's Government:

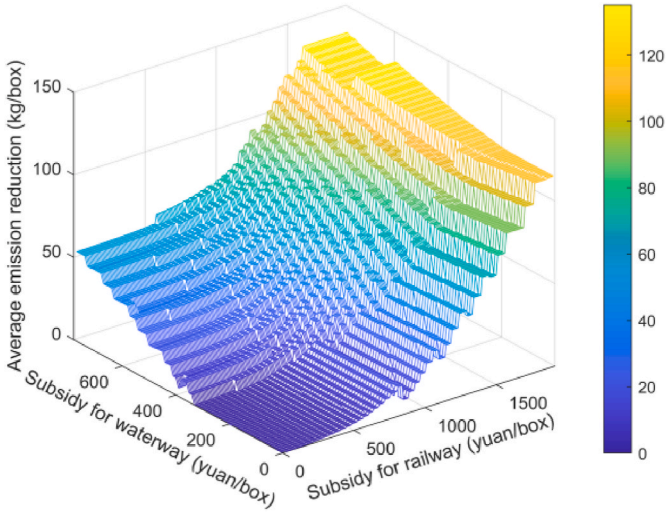


Fig. 4. The relationship between average emission reduction and subsidies.

$$pro_{ship_2025} \geq 52\% \tag{19}$$

$$pro_{rail_2025} \geq pro_{rail_2022} * (1 + 15\%)^3 \tag{20}$$

If the constraints are not all satisfied, F is set to zero in order to find the optimal subsidy policy that satisfies all constraints, as shown in Fig. 5.

To meet the development requirements of the Shanghai government for multimodal transport, the optimal subsidy policy is to subsidize 1220 yuan per container for the railway carrier and 770 yuan per container for the waterway carrier. In this case, the proportions of road, railway, and waterway transport being selected are 31%, 14%, and 55% respectively, and each ton of carbon emission reduction requires a subsidy of 6353 yuan. The current subsidy from the Shanghai government for the railway carrier is 600 yuan per container, which indicates that the Shanghai government needs to increase the subsidies to the railway carrier. Subsidies to the waterway carrier are also indispensable to keep the ratio of the river-sea multimodal transport at no less than 52%.

It is important to note that the model does not necessarily have a solution when the constraints are strict. In addition to providing subsidies, governments can also take measures such as building multimodal

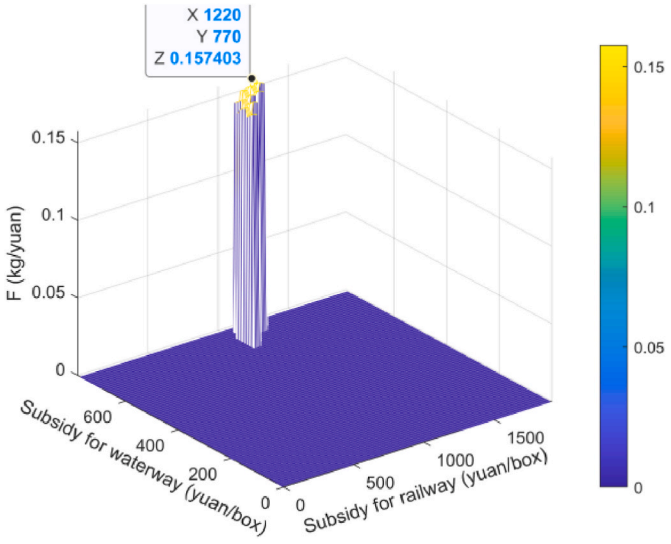


Fig. 5. The relationship between F and subsidies with two constraints.

infrastructure and streamlining the processes of multimodal transport. The continuous provision of large subsidies will create a significant financial burden for the government. To solve this problem, the model can also take the government's emission reduction targets and subsidy capacity into account. Using the same method, the minimum total emission reduction and the maximum subsidies can also be analyzed as constraints to meet the actual needs as long as the total transport volume is known.

4.5. Sensitivity analysis

To investigate the generality of this model, some parameters are changed to analyze their effects on the model. The small number of solutions that satisfy both constraints at the same time is not conducive to sensitivity analysis, so in all subsequent analyses, only the second restriction is added.

(1) Time value of goods (*tv*)

Time cost mainly consists of the time value of goods (*tv*) and the demurrage charge (*dc*). *tv*, which is much larger, may have a greater impact on the calculation results. At the same time, *tv* differs from port to port. For example, the near-ocean export goods to Waigaoqiao has higher timeliness requirement, which is taken as 30 yuan per hour (Huang, 2016), twice as much as that of Shanghai Yangshan Port. In addition, this study does not consider the type of goods. When *tv* is changed, whether the model is still effective needs to be verified. The relationship between *F* and the subsidies when $tv = 15$ and $tv = 30$ is shown in Fig. 6.

Also listed are the relationship between price reduction ratio (*D*), selection proportion (*pro*) of the three modes of transportation and *F* when the optimal subsidy strategy is adopted, as shown in Table 4.

The optimal government subsidy strategy will change significantly with *tv*. Since the proportion of railway transport is small, government subsidies to the railway carrier are more effective in reducing emissions and can promote the development of sea-rail multimodal transport regardless of *tv*. When *tv* becomes larger, more shippers will choose road transport, and the proportion of both rail and waterway being chosen will be reduced. This makes the total number of containers transported by rail and waterway lower, and government subsidies to railway and waterway carriers will achieve better emission reduction. Therefore, statistical analysis of the time value of goods at the port is very important for the government to develop a subsidy strategy. The government can also provide different subsidies for goods with different time values to further improve emission reduction.

Table 4
Optimal subsidy strategy under different time value of goods.

tv (yuan/h)	S (yuan/FEU)	D	pro	F (kg/yuan)
15	[0 1060 0]	[0.20 0.36 0]	[0.50 0.18 0.32]	0.17
30	[0 430 0]	[0.01 0.20 0]	[0.81 0.09 0.10]	0.51

(2) Resilience of shipping

In the selected case, the demurrage charge is a relatively small part of the time cost, since most of the carriers can deliver the containers to their destinations within their free time. However, resilience is still an important factor. Shipping has longer transit times and greater risk of delays than other modes of transportation. The less resilience shipping has, the greater its risk of delay.

Due to the normal distribution of voyage times for large volumes, the risk of delay increases with the standard deviation of voyage times. As shown in Fig. 7, when the government picks the optimal subsidy policy and waterway transport is perfectly on time, the proportion of waterway transport being selected is 43.9%. As the standard deviation of shipping time increases, the proportion of waterway transport being selected

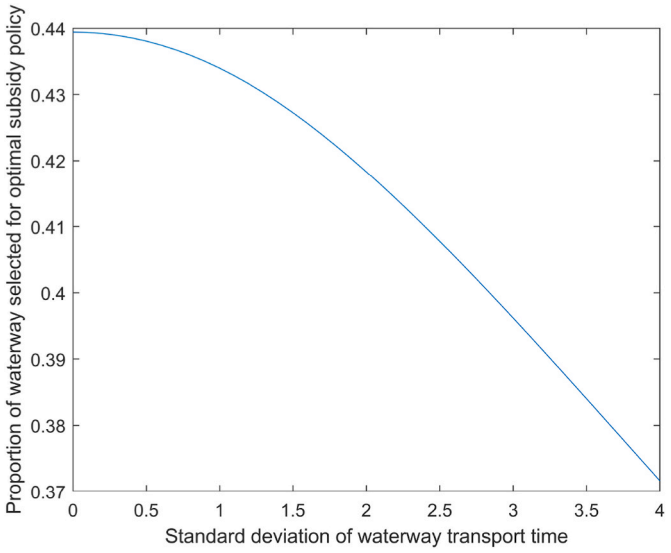
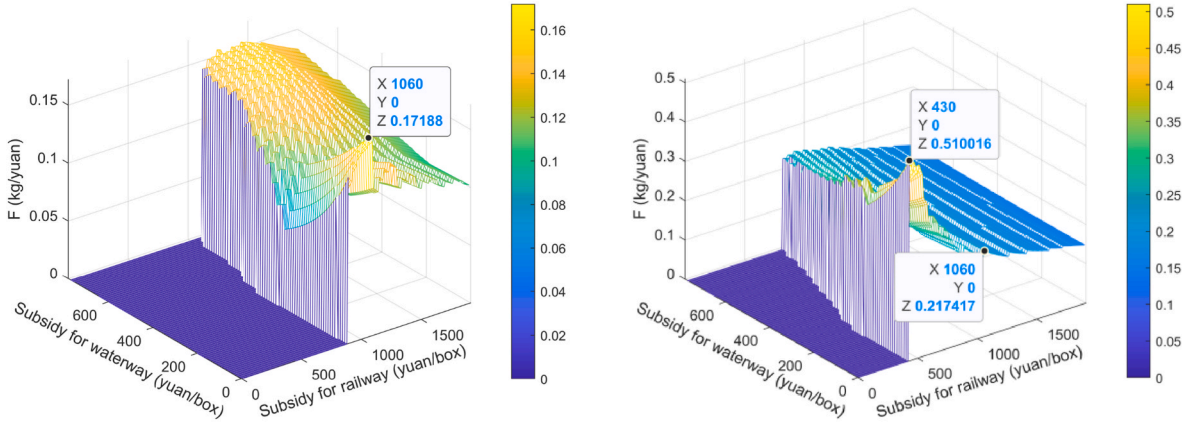


Fig. 7. The relationship between σ and proportion of waterway transport selected.



(a) $tv = 15$ (b) $tv = 30$

Fig. 6. The relationship between *F* and subsidies.

gradually decreases to 37.2%, indicating that the elasticity of shipping can significantly affect shippers' decisions and carriers' profitability.

Fig. 8 illustrates how waterway carrier profits decrease with the variance of shipping time when the government picks the optimal subsidy policy. The profit is obtained by multiplying the profitability of waterway transport by the proportion of it being selected. To calculate the actual profit, it is still necessary to multiply by the total volume of all three modes of transport. Chen et al. (2018) claim that frequent vessel delays can be extremely damaging to market demand, market share, corporate profits and customer loyalty, and may lead to more customers switching to other modes of transport that offer a higher level of service, which is supported by this study.

To avoid this situation, the waterway carrier can learn from the five-scheduled train operated by the railway carrier, thus fixing the time of delivery as much as possible. On the other hand, all parties involved should also actively reduce the cumbersome processes involved in multimodal transport and improve transport resilience to reduce the risk of delay.

In the case of less resilient shipping, the need for government subsidies is even greater for river-sea multimodal transport. At this time, the maximum efficiency of government subsidies is also higher, as shown in Fig. 9. This is because the initial proportion of those choosing waterway transport in the absence of government subsidies falls compared to when shipping is more resilient and rises even more after government subsidies are provided.

(3) Carrier's profit margin (r)

Since carriers cannot operate at a loss and the objective of the game is to maximize profit, r determines their maximum price reduction ratio (D_{max}) and has a direct impact on D. Depending on the market situation of container transport, the carrier profit fluctuates greatly. For example, the overall profit of the logistics industry has decreased in recent years due to COVID-19. Therefore, a sensitivity analysis is performed. Substitute $r = 0.9*r_0$ and $r = 1.1*r_0$ into the model respectively, and analyze the optimal government subsidy strategy when the logistics market sentiment is worse and better, and core the relationship between F and the subsidy amount, as shown in Fig. 10.

Table 5 presents the relationship among the price reduction ratio (D), selection proportion (pro) of the three modes of transportation, and F when the optimal subsidy strategy is adopted.

The model calculation results are similar to the original. When r decreases by 10%, government subsidies for sea-rail transport should be slightly increased to achieve the same reduction effect as the original due to poor market conditions. And when r increases by 10%, government subsidies for sea-rail transport should be slightly reduced, which is also consistent with the reality. When market conditions are worse, carriers need more government subsidies, even if they are slightly less effective in reducing emissions. As highway transport has a higher profit margin, when market conditions become better, highway carriers will have more scope to reduce their prices and will be more attractive to shippers. The efficiency of government subsidies will be higher at this time.

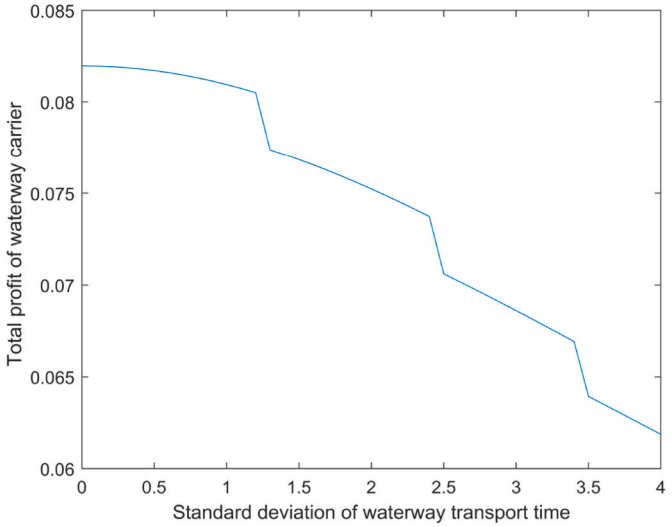


Fig. 8. The relationship between σ and the profit of waterway carrier.

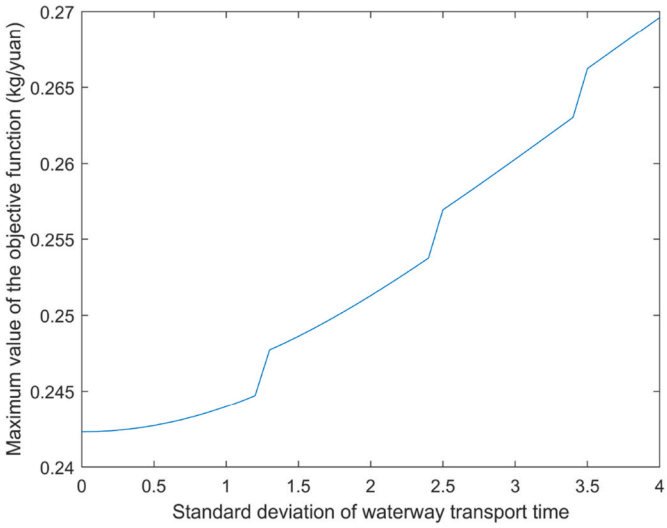
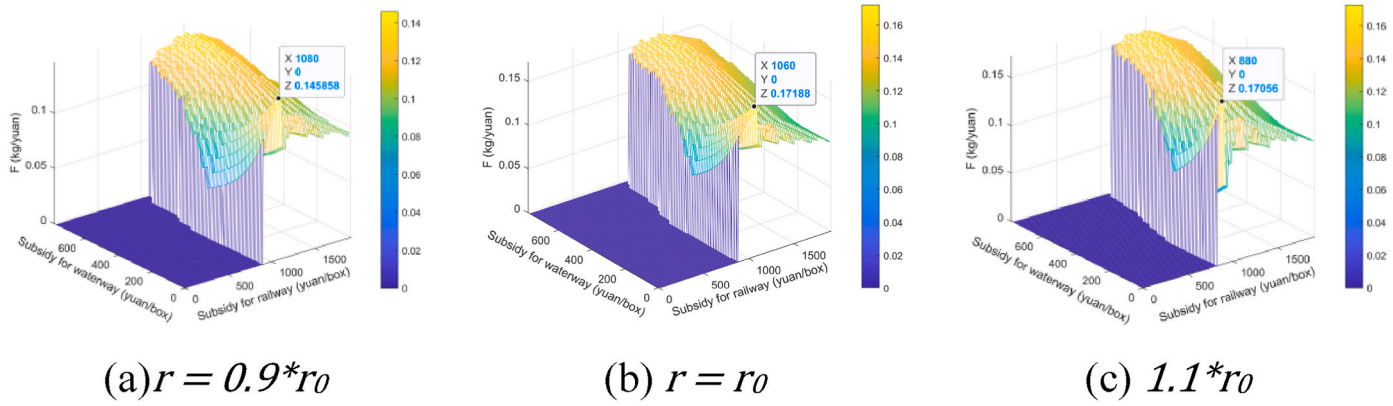


Fig. 9. The relationship between σ and maximum value of the objective function.

By analyzing the game between ports participating in multimodal transport, Jiang et al. (2020) found that the inland transportation cost and transport time of multimodal transport is more important compared to inland transportation. This study corroborates this view and illustrates through quantitative calculations that both the time cost and operating profit of multimodal transport have an impact on government subsidy policies. Therefore, the government should pay attention to these two important indicators to formulate a more scientific and reasonable multimodal transport subsidy policy.

5. Conclusion

To develop scientific and effective multimodal transport subsidy policies to promote the development of multimodal transport, save energy, and reduce carbon emissions, this study establishes a bi-level subsidy decision-making model considering the pricing game of the carriers of three transportation modes. In the upper tripartite game model, the carrier's pricing decision is obtained with the graphical method. The determination of the carrier decision also relies on the selection proportion of three transportation modes obtained by applying the lower logit model. An activity-based method is used to calculate the emission reduction corresponding to the subsidy, to obtain the optimal government subsidy strategy. This study employs the multimodal transport of containers from Nanjing to Shanghai Yangshan port for export as an example. After research interviews and literature studies, data are substituted to verify the robustness, usability, and practicality of the model. To reduce one ton of carbon emissions, 800.7 yuan is required. But in this scheme, the proportion of sea-rail multimodal transport is too low, only 0.27%, which does not achieve the purpose of promoting the development of multimodal transport. According to the development goals of multimodal transport of Shanghai port, two constraints are added. In this case, the best subsidy policy is to subsidize 1220 yuan per container for the railway carrier and 770 yuan per



(a) $r = 0.9*r_0$ (b) $r = r_0$ (c) $1.1*r_0$

Fig. 10. The relationship between *F* and subsidy.

Table 5
Optimal subsidy strategy under different carrier profit.

r	S(yuan/FEU)	D	pro	F(kg/yuan)
0.9*[0.346 0.148 0.042]	[0 1080 0]	[0.18 0.35 0]	[0.44 0.18 0.38]	0.146
[0.346 0.148 0.042]	[0 1060 0]	[0.20 0.36 0]	[0.50 0.18 0.32]	0.172
1.1*[0.346 0.148 0.042]	[0 870 0]	[0.21 0.33 0]	[0.58 0.11 0.31]	0.173

container for the waterway carrier. At this time, the subsidies needed to reduce one ton of carbon emissions are 6353 yuan, while the proportion of sea-rail multimodal transport and river-sea multimodal transport will reach 14% and 55%, respectively.

A sensitivity analysis was performed to analyze the influence of several factors, including the time value of goods, the resilience of shipping, and carrier profits, on the subsidy decision. The results reveal that as the delay risk of water transport increases, the proportion of waterway transport being selected and the operating profit of water carriers will decrease accordingly. Therefore, the waterway carrier should pay more attention to the resilience of shipping and improve the ability to recover from unforeseen circumstances to reduce the risk of delay. Additionally, the government should concern about the time value of goods and the market situation of container transportation to formulate a scientific and effective subsidy policy. For example, because of the higher time value of the cargo at Waigaoqiao Port compared to Shanghai Yangshan Port, subsidizing Waigaoqiao Port is more effective. When the carriers' profit decreases, the government should increase the subsidy appropriately, and vice versa, the subsidy should be reduced appropriately. This quantitative study of multimodal transport subsidy policy innovatively considers the pricing game among highway, railway, and waterway carriers, and proposes the optimal subsidy scheme in combination with actual cases. These findings are instructive for future research on pricing games among carriers in related fields.

Although the case selected in this study is relatively simple, the proposed model is applicable to more complex multimodal transport scenarios with competition among different carriers. It is noteworthy that due to differences in transportation distances and infrastructures such as stations, transit stations, and ports, altering the origin and destination of multimodal transportation will make the compensation cost (A) for the safety and convenience of the transportation scheme incompatible with the values utilized in this study. Accordingly, it needs to be adjusted based on the selection proportion of the actual transportation mode. Furthermore, this study transforms the pricing game into a finite strategy game and employs the graphical method to solve the Nash equilibrium of the tripartite game model. Although this is a simple and effective method, infinite strategy games are more realistic than finite ones. In future research, the solution of the Nash equilibrium of the tripartite game model can be further improved.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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